

Fields of Visibility

Mapping Aesthetic Communities on Instagram Reels

Georgii Bezridnyi

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1 Introduction

This paper argues that one of the primary regulatory strategies at the creators' disposal is aesthetic positioning. Such dynamics have previously been studied on Instagram, a major site of new media cultural production (Abidin 2015; Marwick 2015; Mears 2023). While existing research focuses on the general representation of the creators, comparatively less attention has been paid to the aesthetic organization of the video content itself. Moreover, in recent years, the platform introduced the Reels feature, a short-form vertical videos format, which is primarily driven by algorithmic distribution and is functioning as a competitive arena for visibility. However, this format has received little attention in the studies of the cultural economy of social media. This research aims to address this gap, arguing that the aesthetic choices in Instagram Reels act as a mechanism through which creators mediate between cultural expression and economic pressure.

This research, therefore, asks: How is the field of cultural production in Instagram Reels structured, and how do aesthetic communities emerge among creators? (Goldhaber 1997)

What aesthetic communities can be identified? Is the aesthetic field organized into discrete communities, or does it form continuous categories? How is the thematic difference translated into position within the field?

2 Literature Review

The rise of social media has significantly reshaped both traditional and contemporary forms of cultural production. A particularly substantial novelty is social media adoption of the "platform" model by providing a technical means (e.g., APIs) for various actors to reprogram the service for their needs (Gillespie 2010). The active "*platformization*" enabled a collective economic activity, essentially creating multi-sided markets that connect different participants (Helmond 2015). It is important to note that the platforms themselves act as the stakeholders and active participants in these economic relations.

Besides economic implications, many notable scholars mention the intricate power structures. It is often noted that the platform, in fact, has a distinctively strong agency (Helmond 2015; Van Dijck 2013). The manifestations of non-neutrality are constituted individually for each platform and are deeply embedded into a certain design, specific affordances, or algo-

rhythmic interventions (Bucher and Helmond 2017). These material aspects are non-static and are constantly adapted to the needs of the actors, be it the mundane measures serving the market economy, or ways to maintain and interact with users. In this sense, the platform is more than a sole intermediary, as “it shapes the performance of social acts instead of merely facilitating them” (Van Dijck 2013).

A vital component of the ecosystem is content creators, who essentially provide the cultural goods, the user-generated content (UGC). Their practices are shaped by and, in turn, reproduce the infrastructural logic. At the same time, these actors maintain a certain degree of agency to balance their economic and cultural incentives by means of self-presentation (Marwick 2015, Senft 2013). This status quo constitutes the “*attention economy*”, a competitive model, where the primary scarce resource is the audience (Goldhaber 1997).

To sum up briefly, social media are characterized by a complex network of both formal and implicit relations between different categories of actors. This makes the analysis through a cultural and economic perspective particularly effective. In this context, the platform can be conceptualized as a field of cultural production, which invites the application of Pierre Bourdieu’s (1993) theoretical agenda. According to Bourdieu, a field is a structured space in which actors compete for recognition and resources in the form of economic and symbolic capital. Positions within the field are fundamentally relational and dynamic, as they are determined by varying strategies, allocation of resources, and wavering legitimacy. The internal structure of the field is organized around differently oriented poles - towards symbolic capital and cultural value or towards economic capital and external demand. Belonging to a

certain pole is determined by stylistic choices and practices made by cultural producers.

This paper largely leans on computational semantics as its primary toolkit. One of the key analysis principles is the latent representation. Deriving from linguistics, the Latent Semantic Analysis suggests that: “*the aggregate of all the word contexts in which a given word does and does not appear ... determines the similarity of meaning of words and sets of words to each other*” (Landauer, Foltz, and Laham 1998). In essence, it was proposed to approach the language not as a static set of discrete units, but rather as a relational structure. Meanings in such a system occur from contextual patterns instead of predetermined characteristics. LSA became particularly prominent in its discipline and rapidly spread to computational fields. For instance, computer science rapidly adopted this structural approach by operating representations in the form of mathematically determined objects. This distinctive formalization allows to represent meanings in a coordinated vector space, where the semantics of an entity is coined as embedding. A principle is canonically described by the pioneer researchers in a classical example: “*vector(‘King’) – vector(‘Man’) + vector(‘Woman’) results in a vector that is closest to the vector representation of the word ‘Queen’*” (Mikolov et al. 2013). This approach is increasingly applied outside of linguistic data, featuring works in digital humanities and cultural analytics. Noting that cultural data is constructed and prone to bias, Lev Manovich (2020) calls for this approach: “*Rather than starting with existing and seemingly natural categories, the cultural analytics approach is to extract low-level features and find clusters of objects based on these features.*” Therefore, this paper aims to utilize the high-dimensional semantic space to organize the cultural data in a measurable form. Such a def-

inition enables the quantification of relations between the entities by measuring their geometric proximity. This technique provides a clear operationalization for the central concept of this paper, the shared aesthetic.

3 Theoretical Framework

Based on Bourdieu’s theory (1993), this study conceptualizes Instagram Reels as a space of cultural production. Within the field, content creators can be understood as microcelebrities incentivized by attention and visibility, who cultivate their audience through strategic self-presentation and stylistic choices (Senft 2013). Such behavior demonstrates their agency, as they maneuver between cultural expression and economic incentives. Essentially, aesthetic choices act as a positioning mechanism. Previous research has shown that such visibility systems tend to reproduce established symbolic conventions. (Marwick 2015). These conventions act like a shared aesthetic repertoire, an organizing principle for the whole field. It plays a structural role to connect creators with the audience and giving rise to distinct communities within Instagram Reels. Consequently, this study understands aesthetics not as predefined categories, but rather as relational patterns deriving from similarities in content and attributing to a related aesthetic community.

4 Methodology

To empirically investigate these relations, computational semantics methods are employed. Specifically, this paper aims to represent videos in a high-dimensional vector space. This approach allows to cluster stylistically similar content, respectively, highlighting various aesthetic communities (Martin 2017).

The manually compiled dataset consists of various Instagram accounts, a sample of 1 000

content creators who meet certain criteria. Candidates have more than 10 000 followers to account for niche creators. They have posted more than 10 new Reels in the last 180 days. Demographics are a particularly keen and impactful aspect of this research as it influences which the communities emerge. As study aims to provide a general definition of how Reels function as a cultural field, a broad sample will be used, comprising Western accounts that use English as their primary language. Importantly, the dataset is heterogeneous and represent different thematic categories (beauty, sports, music, politics, etc.). Such dataset design is prone to selection bias it and will be considered in the analysis.

For each creator, the 5 most viewed Reels posted in the last 180 days are examined. Keyframes, audio, and caption features are extracted. Each modality is to be converted into text using open weights, providing transparency and control. (Li et al. 2026) The combined information is used to obtain an embedding of a video. The average vector of five embeddings is a representation of the creator. A control methodological aspect of the study involves experimenting with modalities for final vectorization. Depending on selected domains, the resulting communities may differ significantly. The study attempts to approximate the natural consumption experience, therefore, the result of all modalities is conceptually preferred, though other implications will be considered during the analysis.

Finally, for effective clustering, the high-dimensional vectors should be reduced using UMAP, a contemporary dimensionality reduction technique (McInnes et al. 2018). The further clustering is done via HDBSCAN, a density-based clustering algorithm without a predefined number of clusters (McInnes, Healy, and Astels 2017). For further visual analysis, a

reduction to 2D should be performed, as well as the color-coding of clusters and belonging vertices.

The resulting clusters represent aesthetic communities, populated by creators with similar aesthetic features. Both the internal structure of clusters and the overall organization of the map will be examined to characterize the field. Descriptive analysis will provide answers to the research questions posed.

5 Ethical considerations

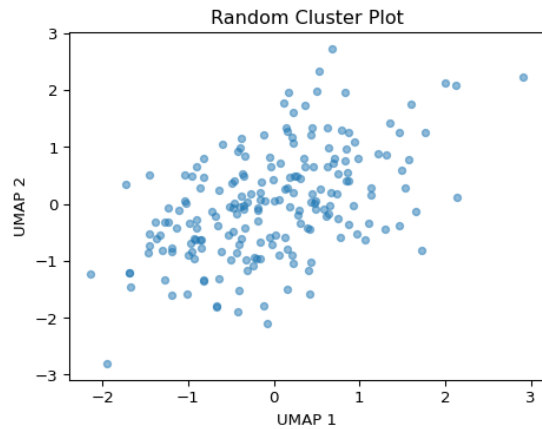
The study relies on publicly available user data from Instagram. The data consists of the following data types: usernames, profile descriptions, locations, publicly posted user-generated video content (that may include identifiable faces and voices).

6 Expected findings

The analysis is expected to reveal a multitude of aesthetic communities that correlate with recognizable aesthetic categories. The overall structure of the embedding space is expected to show both clearly defined clusters and broader ones. It should be possible to see transitional cultural producers, for example, due to thematic and stylistic inconsistencies. Within clusters, points gravitate in different directions depending on the selected visual features, music, and description.

From the perspective of fields theory, clusters correspond to the logic of orientation according to strategies prioritizing different cultural and economic capitals. Creators with similar strategies form a community. Some of them prefer mass engagement, while others form cultural niches. The resulting map will be an empirical representation of the relational structure of the field.

7 Results



Bibliography

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